

Association Learning in Unsupervised Learning

Introduction

Association Learning is a major branch of **Unsupervised Learning** in Machine Learning. It is used to discover interesting relationships, patterns, and associations between variables in large datasets. Unlike supervised learning, association learning works without labeled outputs.

The main goal is to identify rules that describe how items are associated with each other in transactions or datasets.

Example:

- Customers who buy **bread** often also buy **butter**.
- Customers who purchase **milk** may also purchase **cereal**.

This type of analysis is widely used in:

- Market Basket Analysis
 - Recommendation Systems
 - Web Usage Mining
 - Medical Diagnosis
 - Fraud Detection
 - Inventory Management
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What is Association Rule Learning?

Association Rule Learning finds relationships among items in transactional datasets.

It works using rules in the form:

$$X \Rightarrow Y$$

Where:

- X = antecedent (input item set)
- Y = consequent (result item set)

Meaning:

If a customer buys X , they are likely to buy Y .

Example:

$$\{Bread\} \Rightarrow \{Butter\}$$

Meaning customers buying bread are likely to buy butter.

Key Terminologies

1. Support

Support measures how frequently an itemset appears in the dataset.

Formula:

$$Support(X) = \frac{Transactions\ containing\ X}{Total\ Transactions}$$

Higher support means the item appears more frequently.

2. Confidence

Confidence measures the probability that item Y is purchased when item X is purchased.

Formula:

$$Confidence(X \Rightarrow Y) = \frac{Support(X \cup Y)}{Support(X)}$$

Higher confidence means stronger association.

3. Lift

Lift measures how strongly two items are associated compared to random occurrence.

Formula:

$$Lift(X \Rightarrow Y) = \frac{Confidence(X \Rightarrow Y)}{Support(Y)}$$

Interpretation:

- Lift > 1 → Positive association
 - Lift = 1 → Independent items
 - Lift < 1 → Negative association
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Types of Association Learning Algorithms

The most commonly used algorithms are:

1. Apriori Algorithm
 2. FP-Growth (Frequent Pattern Growth) Algorithm
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Apriori Algorithm

Introduction

The Apriori Algorithm is one of the most popular algorithms used for association rule mining and frequent itemset mining.

It works on the principle:

“If an itemset is frequent, then all of its subsets must also be frequent.”

This principle is called the **Apriori Property**.

Working of Apriori Algorithm

The algorithm works in the following steps:

Step 1: Find Frequent Individual Items

Calculate support for each item.

Remove items below minimum support threshold.

Step 2: Generate Candidate Itemsets

Combine frequent items to create larger itemsets.

Example:

- {Bread}
- {Butter}

Combine to:

- {Bread, Butter}
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Step 3: Calculate Support

Find support of generated itemsets.

Remove itemsets below minimum support.

Step 4: Repeat

Continue generating larger itemsets until no more frequent itemsets are found.

Step 5: Generate Association Rules

Generate rules using:

- Confidence
 - Lift
 - Support
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Apriori Algorithm Implementation

Libraries Used

The notebook implementation used:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Additional library commonly used:

```
from mlxtend.frequent_patterns import apriori, association_rules
```

Dataset Preparation

The grocery dataset is loaded:

```
df = pd.read_csv("groceries1.csv")
```

The dataset contains transactional records.

Data Preprocessing

Transactions are converted into list format:

```
market=[]
for i in range(0,df.shape[0]):
    cus=[]
    for j in df.columns:
        if type(df[j][i])==str:
            cus.append(df[j][i])
    market.append(cus)
```

This creates a list of customer transactions.

Applying Apriori

Example implementation:

```
frequent_items = apriori(dataframe,
                          min_support=0.02,
                          use_colnames=True)
```

Generating Rules

```
rules = association_rules(frequent_items,
                          metric="confidence",
                          min_threshold=0.5)
```

Advantages of Apriori Algorithm

- Easy to understand
 - Simple implementation
 - Effective for small datasets
 - Generates interpretable rules
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Disadvantages of Apriori Algorithm

- Slow for large datasets
 - Generates many candidate itemsets
 - Requires multiple database scans
 - High memory consumption
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FP-Growth (Frequent Pattern Growth) Algorithm

Introduction

FP-Growth is an improved algorithm over Apriori.

It avoids generating candidate itemsets explicitly.

Instead, it uses a special tree structure called the:

FP-Tree (Frequent Pattern Tree)

This makes FP-Growth significantly faster and more memory efficient.

Working of FP-Growth Algorithm

Step 1: Scan Dataset

Find frequency of all items.

Remove infrequent items.

Step 2: Construct FP-Tree

Build a compressed tree structure representing transactions.

Common prefixes are shared.

Step 3: Generate Conditional Pattern Base

Extract conditional patterns for each item.

Step 4: Generate Frequent Patterns

Mine frequent itemsets recursively.

FP-Growth Implementation

Libraries Used

```
from mlxtend.frequent_patterns import fpgrowth
```

Applying FP-Growth

```
frequent_patterns = fpgrowth(dataframe,  
                             min_support=0.02,  
                             use_colnames=True)
```

Generating Association Rules

```
rules = association_rules(frequent_patterns,  
                          metric="confidence",  
                          min_threshold=0.5)
```

Advantages of FP-Growth

- Faster than Apriori
 - No candidate generation
 - Efficient for large datasets
 - Requires fewer database scans
 - Lower memory usage
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Disadvantages of FP-Growth

- Complex implementation
 - FP-tree construction can be difficult
 - Less intuitive compared to Apriori
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Difference Between Apriori and FP-Growth

Feature	Apriori	FP-Growth
Candidate Generation	Yes	No
Speed	Slower	Faster
Memory Usage	High	Lower
Database Scans	Multiple	Fewer
Complexity	Simple	Complex
Efficiency on Large Data	Low	High

Real-World Applications

1. Market Basket Analysis

Retail stores analyze customer purchasing behavior.

Example:

- Amazon product recommendations
 - Supermarket product placement
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2. Recommendation Systems

Streaming platforms and e-commerce websites recommend related items.

Examples:

- Netflix movie recommendations
 - YouTube suggestions
 - Online shopping recommendations
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3. Healthcare

Detect relationships between symptoms and diseases.

4. Fraud Detection

Identify suspicious transaction patterns.

5. Web Usage Mining

Analyze user navigation behavior on websites.

Challenges in Association Learning

- Large search space
 - High computational cost
 - Redundant rules
 - Choosing optimal support and confidence
 - Handling sparse datasets
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Conclusion

Association Learning is a powerful unsupervised learning technique used to discover hidden relationships in data.

The two major algorithms are:

1. Apriori Algorithm

2. FP-Growth Algorithm

Apriori is simple and easy to understand but slower for large datasets.

FP-Growth is faster and more efficient because it avoids candidate generation using FP-Trees.

These algorithms are widely used in:

- Retail analytics
- Recommendation systems
- Healthcare
- Fraud detection
- Customer behavior analysis

Association Learning plays an important role in data mining and business intelligence by helping organizations make data-driven decisions.

Short Summary

- Association Learning is an unsupervised learning technique.
- It discovers relationships among items.
- Important metrics:
 - Support
 - Confidence
 - Lift
- Apriori uses candidate generation.
- FP-Growth uses FP-Tree.
- FP-Growth is faster than Apriori.
- Common application: Market Basket Analysis.